Topic 4.1 - Balancing Chemical Equations

Balancing

1. Why does a chemical equation need to be balanced?

2. What does a balanced chemical equation represent? (Think about recipes in cooking)

3. Balance the following equations!

a. ___H_3PO_4 + ___KOH → ___K_3PO_4 + ___H_2O

b. ___K + ___B_2O_3 → ___K_2O + ___B

c. ___HCl + ___NaOH → ___NaCl + ___H_2O

d. ___Na + ___$NaNO_3$ → ___Na_2O + ___N_2

e. ___C + ___S_8 → ___CS_2

f. ___Na + ___O_2 → Na_2O

g. ___H_3PO_4 + ___$Mg(OH)_2$ → ___$Mg_3(PO_4)_2$ + ___H_2O

h. ___N_2 + ___O_2 → ___N_2O_5

i. ___Al + ___S_8 → ___Al_2S_3

j. ___Rb + ___RbNO → ___Rb_2O + ___N_2

k. ___C_2H_6 + ___O_2 → ___CO_2 + ___H_2O

l. ___NH_3 + ___O_2 → ___N_2 + ___H_2O

m. ___FeS_2 + ___O_2 → ___Fe_2O_3 + ___SO_2

4.1 Part 2 - Writing Chemical Reactions

Write the balanced chemical equation for the following reactions.

1. Sodium metal reacts with chlorine gas to produce sodium chloride.

2. Aluminum metal reacts with aqueous copper (II) nitrate to produce aqueous aluminum nitrate and copper metal.

3. Aqueous Nitric acid reacts with aqueous magnesium hydroxide to produce aqueous magnesium nitrate and liquid water.

4. Propane gas (C_3H_8) reacts with oxygen gas to produce gaseous carbon dioxide and gaseous water.

5. Solid sodium carbonate decomposes into solid sodium oxide and carbon dioxide gas.

6. Solid ammonium chloride decomposes into gaseous nitrogen trihydride and gaseous hydrochloric acid.

7. Aqueous lead(II) nitrate and aqueous potassium iodide react to produce solid lead(II) iodide and aqueous potassium nitrate.

Turn the following chemical equations into word equations

8. $Na_2O\,(s) + H_2O\,(l) \rightarrow 2NaOH\,(aq)$

9. $CS_2\,(l) + 3O_2\,(g) \rightarrow CO_2\,(g) + 2SO_2\,(g)$

10. $K\,(s) + CuNO_3\,(aq) \rightarrow KNO_3\,(aq) + Cu\,(s)$

11. $2Li\,(s) + 2H_2O\,(l) \rightarrow 2LiOH\,(aq) + H_2\,(g)$

Topic 4.2 - Reaction Types
Give the reaction type for the following:

1. $C_5H_{12,}$ + $8O_2$ → $5CO_2$ + $6H_2O$ _____
2. $2K_3PO_4$ + $3BaCl_2$ → $Ba_3(PO_4)_2$ + 6KCl _____
3. $2Al(OH)_3$ + $3H_2SO_4$ → $Al_2(SO_4)_3$ + $3H_2O$ _____
4. $2AgNO_3$ + Na_2CO_3 → Ag_2CO_3 + $2NaNO_3$ _____
5. $2La_2O_3$ → 4Al + $3O_2$ _____
6. 2Al + 6HCl → $3H_2$ + $2AlCl_3$ _____
7. $4C_4H_9O$ + $27O_2$ → $18H_2O$ + $20CO_2$ _____
8. 3HBr + $Al(OH)_3$ → $3H_2O$ + $AlBr_3$ _____ _____
9. $2NH_3$ + H_2SO_4 → $(NH_4)_2SO_4$ _____
10. 2HgO → 2Hg + O_2 _____
11. Na_2O + H_2O → 2NaOH _____
12. Cu + $2AgNO_3$ → 2Ag + $Cu(NO_3)_2$ _____

Topic 4.3 - Predicting Products (Synthesis, Decomposition, Combustion)

1. $N_2 + H_2 \rightarrow$ _____
2. $Fe + S_8 \rightarrow$_____
3. $K + Br_2 \rightarrow$_____
4. $NaCl \rightarrow$_____
5. $MgCl_2 \rightarrow$_____
6. $CH_4 + O_2 \rightarrow$_____
7. $H_2O \rightarrow$ _____
8. $Al_2O_3 \rightarrow$_____
9. $C_2H_6 + O_2 \rightarrow$_____
10. $C_{12}H_{26} + O_2 \rightarrow$_____
11. $CH_3OH + O_2 \rightarrow$_____

Topic 4.4 - Predicting Products Single Replacement

1. $Li_{(s)}$ + $BaCl_{2(s)}$→ _____
2. $Li_{(s)}$ + $CaS_{(s)}$→_____
3. $BaCl_{2(s)}$ + $K_{(s)}$→_____
4. $NaF_{(s)}$ + $Ba_{(s)}$→ _____
5. $MgCl_{2(s)}$ + $Na_{(s)}$→_____
6. $NaCl_{(s)}$ + $F_{2(g)}$→_____
7. $NaF_{(s)}$ + $Cl_{2(g)}$→_____
8. $Cu_{(s)}$+ $FeSO_{4(s)}$→ _____
9. $H_2O_{(aq)}$ + $Na_{(s)}$-->_____
10. $H_2O_{(aq)}$ + $Ag_{(s)}$-->_____

Topic 4.5 - Predicting Products Double Replacement

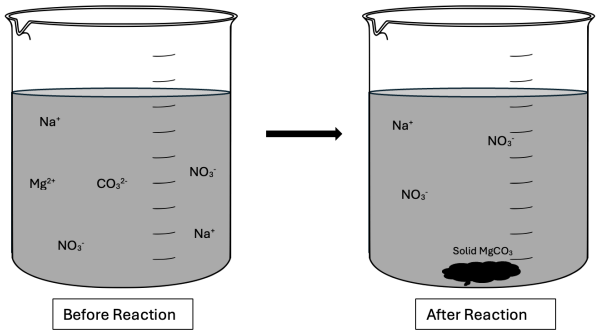
11. $BaCl_{2(aq)}$+ $H_3PO_{4(aq)}$→ _____
12. $K_2SO_{4(aq)}$ + $MgF_{2(aq)}$-->_____
13. $AlCl_{3(aq)}$ + $BaSO_{4(aq)}$-->_____
14. $Li_2S_{(aq)}$ + $AgNO_{3(aq)}$ →_____
15. $BaCl_{2(aq)}$ + $Na_2CO_{3(aq)}$ →_____
16. $Al_2(SO_4)_{3(aq)}$+ $Na_3PO_{4(aq)}$ →_____
17. $Na_2SO_{4(aq)}$+ $NH_4I_{(aq)}$ →_____
18. $Li_3PO_{4(aq)}$+ $BaCl_{2(aq)}$ →_____
19. $AlBr_{3(aq)}$+ $H_3PO_{4(aq)}$ →_____
20. $Zn(NO_3)_{2(aq)}$+ $Ba(OH)_{2(aq)}$ → _____

Topic 4.6 - Net Ionic Equations of Single Replacement and Double Replacement Reactions

1. What does a net ionic equation represent?

2. The species that DO NOT react are called what?

3. A student dissolves sodium carbonate and magnesium nitrate into water. The diagram provided shows before and after the reaction.



   a. Write the balanced chemical equation for the entire reaction.

   b. What ions are reacting and what ions are spectators?
      i. Reacting: _____
      ii. Spectators: _____

   c. Justify your answer to part B using the diagram provided.

   d. Write the net ionic equation for the reaction.

4. Write the net-ionic equation for the reactions below!

   a. $SrBr_{2\,(aq)} + K_2SO_{4\,(aq)} \quad SrSO_{4\,(s)} + 2KBr_{\,(aq)}$

   b. $3Ca(OH)_{2\,(aq)} + 2FeCl_{3\,(aq)} \quad 3CaCl_{2\,(aq)} + 2Fe(OH)_{3\,(s)}$

   c. $2\ AgOH(s) + H_2CO_3(aq) \text{ ---> } Ag_2CO_3(s) + 2\ H_2O(l)$

   d. $Fe(s) + NiCl_3(aq) \text{ ---> } FeCl_3(aq) + Ni(s)$

   e. The reaction between sodium sulfide and iron(II) nitrate

Topic 4.7 - More Naming! (So much fun)

1. Li_2S _____
2. beryllium phosphide _____
3. $Be(OH)_2$ _____
4. P_2Br_4 _____
5. SF_6 _____
6. magnesium acetate _____
7. diboron tetrahydride _____
8. dinitrogen trioxide _____
9. hydrocyanic acid _____
10. lithium iodide _____
11. $KMnO_4$ _____
12. $HClO_3$ _____